COMPUTER NETWORKS

(CSC 258)

(B.Sc. CSIT – II/IV)

Unit-3

Part 2

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UNIT3 – DATA LINK LAYER

3.1. Function of Data Link Layer (DLL)

3.2. Overview of Logical Link Control (LLC) and Media Access Control (MAC)

3.3. Framing and Flow Control Mechanisms

Stop-and-wait ARQ, Piggybacking, Go-Back-N ARQ, Selective Repeat ARQ

3.4. Error Detection and Correction techniques

Parity checks, Cheksumming Methods, CRC, Hamming code

3.5. Channel Allocation Techniques

ALOHA, Slotted ALOHA, CSMA, CSMA/CD, CSMA/CA

3.6. Ethernet Standards → *802.3 CSMA/CD, 802.4 Token Bus, 802.5 Token Ring*

3.7. Wireless LAN → *Spread Spectrum, Bluetooth, Wi-Fi*

3.8. Overview Virtual Circuit Switching, Frame Relay & ATM

3.9. DLL Protocol → *HDLC, PPP*

TYPES OF NETWORK (BASED ON DATA TXN.)

- **Switched Communication Networks**
  - Users are interconnected by the means of transmission lines, multiplexers and switches.
  - Peer to Peer Communication.
  - eg. Telephone Network, X.25, SONET(Synchronous Optical NETwork), Frame Relay, ATM, etc.
- **Broadcast Networks**
  - A single transmission media is shared by all the users and information is broadcast by an user into the medium.
  - eg. Multi Tapped Bus, Ring Network, Satellite Communication Network, Packet Radio Network, Wireless Communication sharing a frequency band, etc.

BROADCAST NETWORK COMMUNICATION

How do the devices in Broadcast Networks communicate?

- **How** is the control exercised?
- **Who** can access the channel & when?

Since, at a time only one device/user can send to the transmission media.

Therefore, Broadcast networks require a protocol to orchestrate (arrange) the transmission from the users.

The protocols used for this purpose are known as **Medium Access Control (MAC) protocols**.

CHANNEL ACCESS MECHANISMS (WHO?)

Centralized Mechanism:

- A designated station has an authority to grant access to the network
- Simple logic at each station
- Greater control to provide feature like priority, overrides and guaranteed bandwidth
- Easy coordination
- Lower reliability – if in case the central controller fails.

Distributed Mechanism:

- Stations can dynamically determine transmission order in a mutual agreement manner.
- Complex, Reliable and Scalable

CHANNEL ACCESS MECHANISMS (HOW?)

Synchronous Manner:

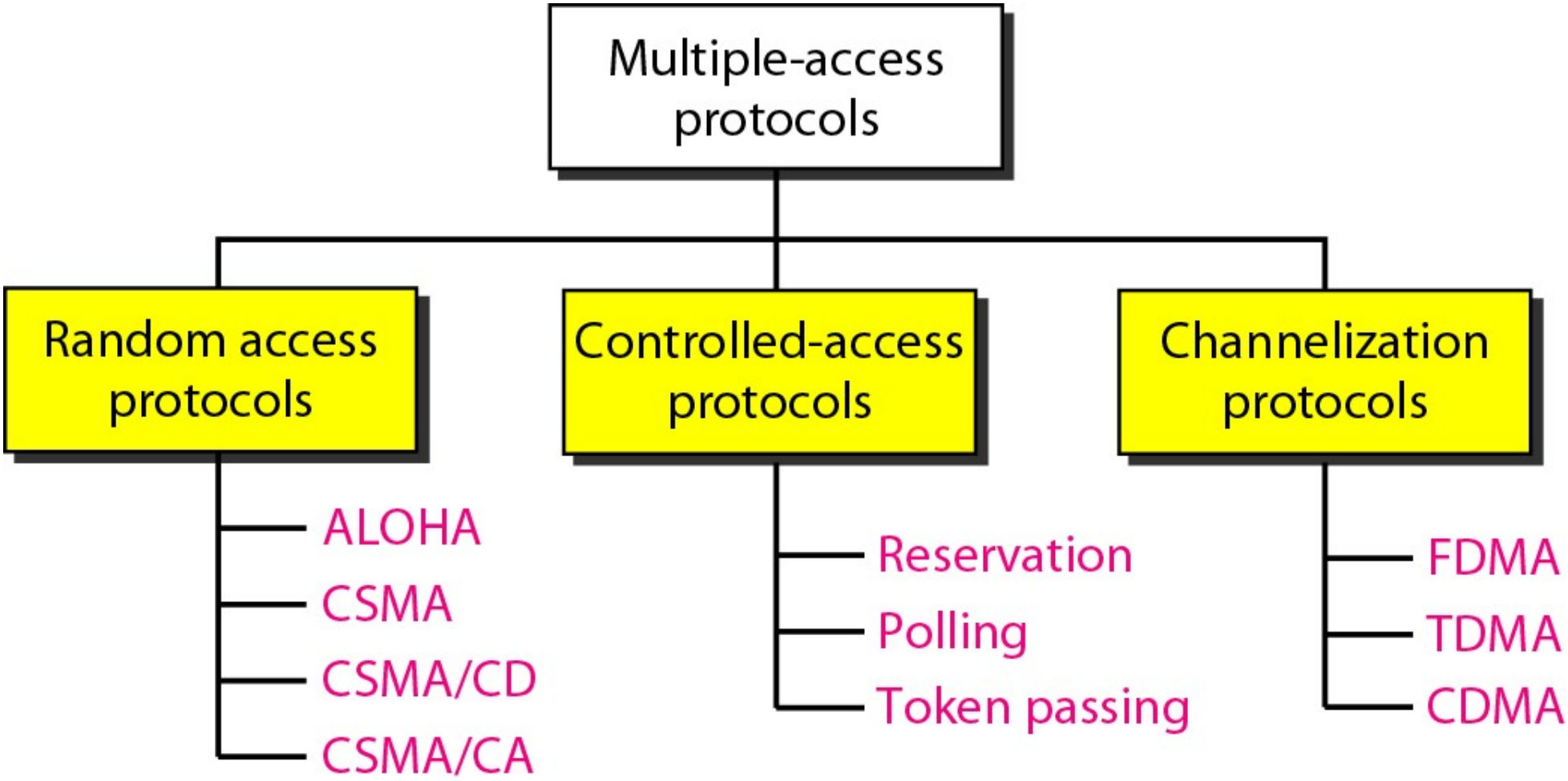
- Dedicated specific capacity to a connection.
- Each slot is allocated to particular user/station.
- eg. TDM, FDM

Asynchronous Manner:

- Capacity is allocated dynamically.
- Most of our data is found in bursty nature thus dedicated capacity may not be used all the time.

GOALS OF MEDIA ACCESS CONTROL(MAC)

- **Initialization** – Stable state after turning on
- **Fairness** – Each node has to be given equal opportunity for sharing
- **Priority** mechanism must also be established
- **Limitation to one station** for transmission
- **Receipt for packets** should be done in orderly manner
- **Error Limitation** – Error detection and correction
- **Reliability** – Should work under certain failure conditions
- **Reconfigurability** – Stations may be added/removed from network
- **Compatibility** – Different vendor's equipment must communicate
- **Recovery** should be there if any failure in network occurs


Multiple-access protocols
Random access protocols
Controlled-access protocols
Channelization protocols
ALOHA
CSMA
CSMA/CD
CSMA/CA
Reservation
Polling
Token passing
FDMA
TDMA
CDMA

MEDIUM ACESS CONTROL TECHNIQUES

Random Access

- ALOHA
  - Pure ALOHA
  - Slotted ALOHA
- CSMA
- CSMA/CD
- CSMA/CA

Controlled-Access Protocols

- Reservation
  - Centralized
  - Distributed
- Round Robin
  - Polling
  - Token Passing

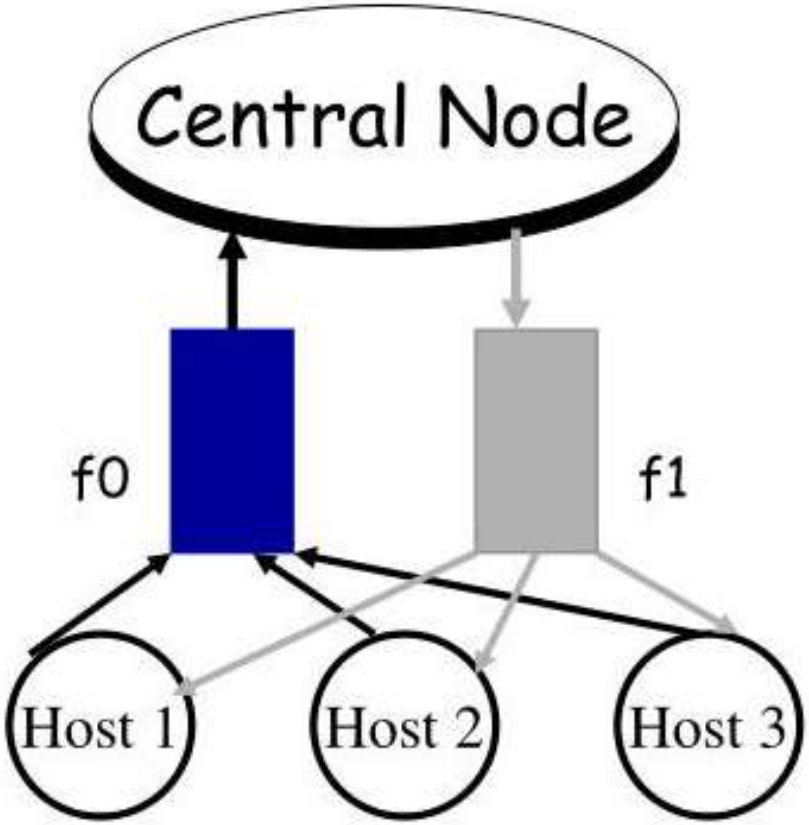
Channelization Protocols

- FDMA
- TDMA
- CDMA

ALOHA

- ALOHA was developed by Norman Abramson at University of Hawaii for the Packet Radio Network; it became operational in June 1971.

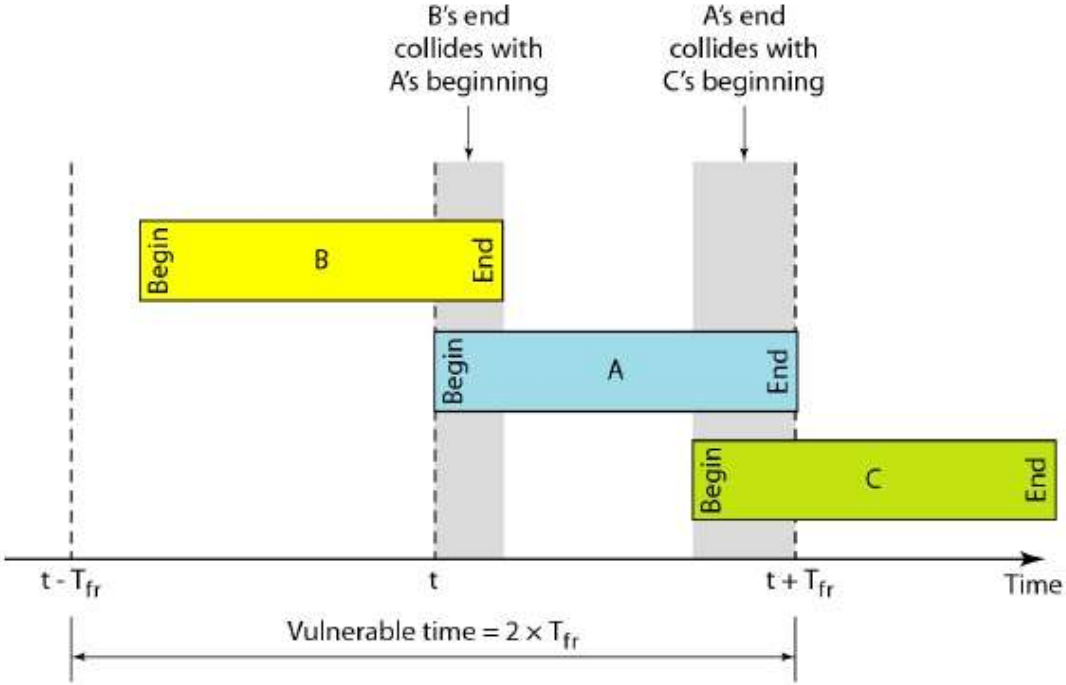
- Hawaiian Island, USA contains a number of islands 137 (8 bigger).
- Connecting them via wired network was a difficult job.
- University of Hawaii had central computer in University and terminals were distributed in different islands.
- It was a LAN based random access protocol
- Frequencies:
  - *f0 → Random Access*
  - *f1 → Broadcast*



ALOHA

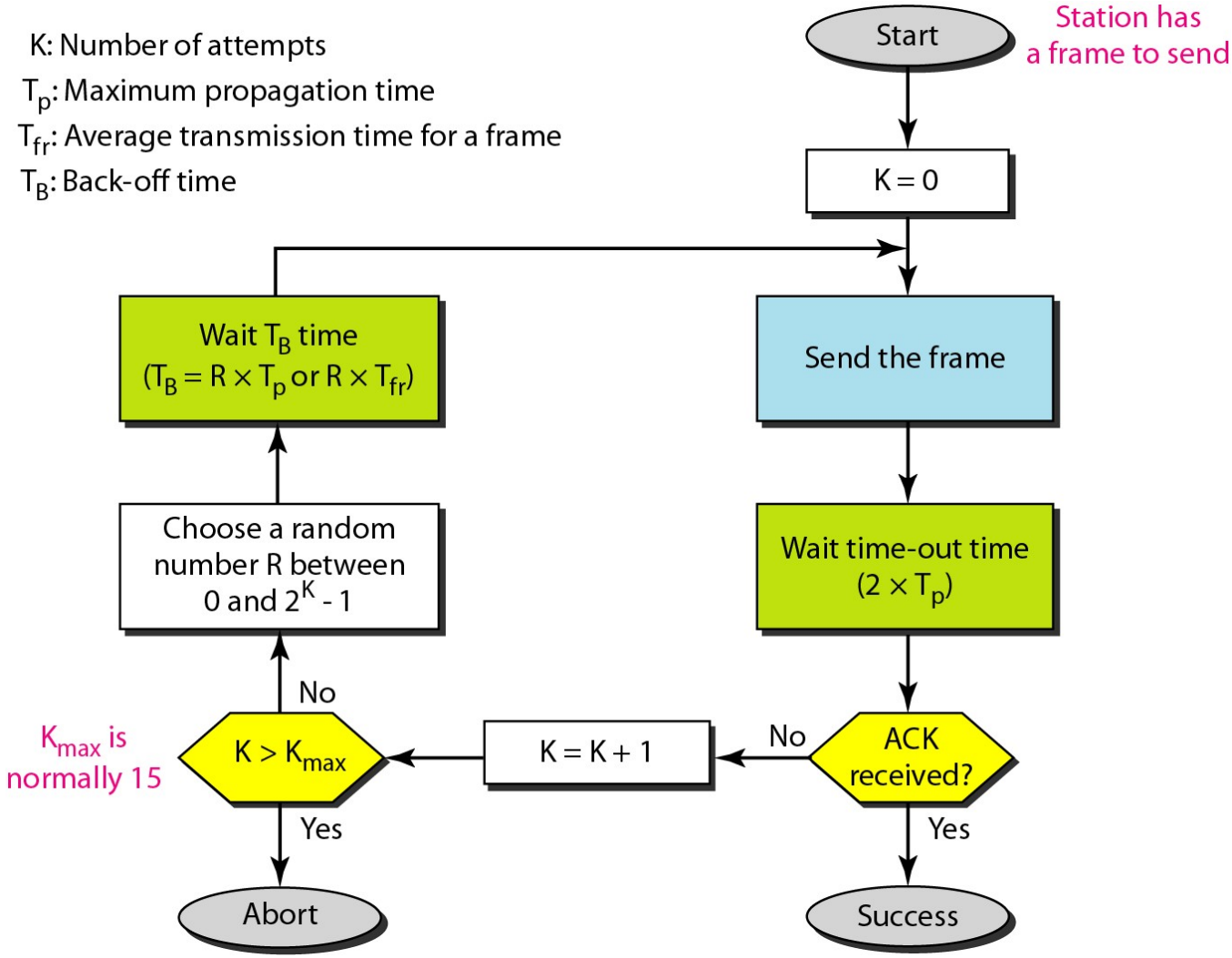
- ***Channel – Free for All***:
- Any station that needs to send data can send at any time via the uplink frequency.
- If no acknowledgement is received, sender assumes that there has been collision.
- Whenever there is collision, stations wait for random time which is called **BACK OFF** period.
- Retransmission is done in the case of collision.

- ***Contention System***: It's the system in which multiple users/stations share a common channel in a way that can lead to conflicts.



T_{fr} – Frame Transmission Time
$2T_{fr}$ – Vulnerable Time Period

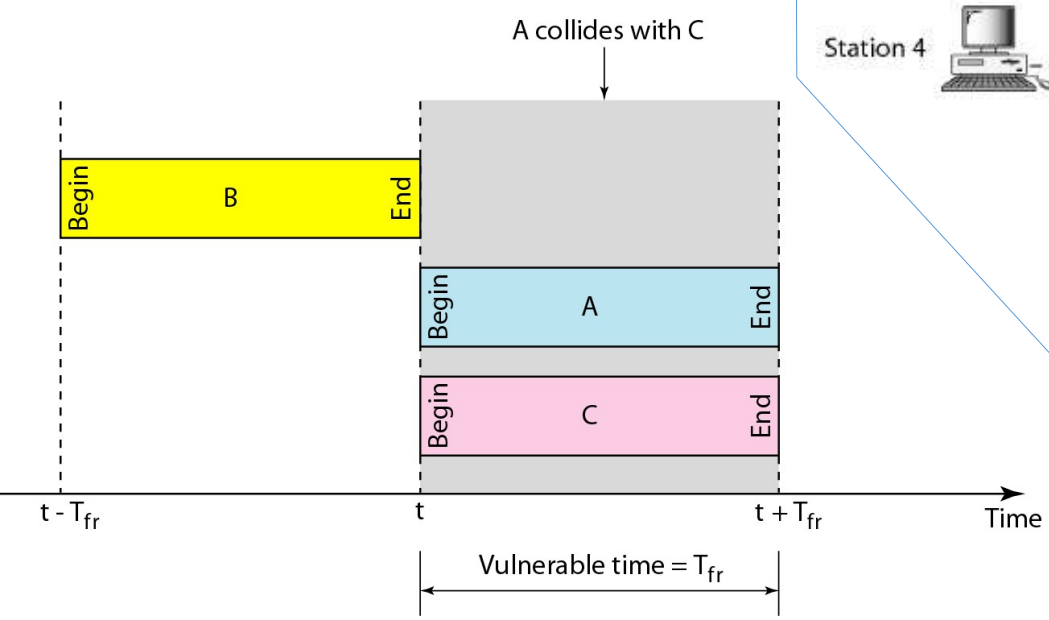
PROCEDURE (ALOHA aka PURE ALOHA)

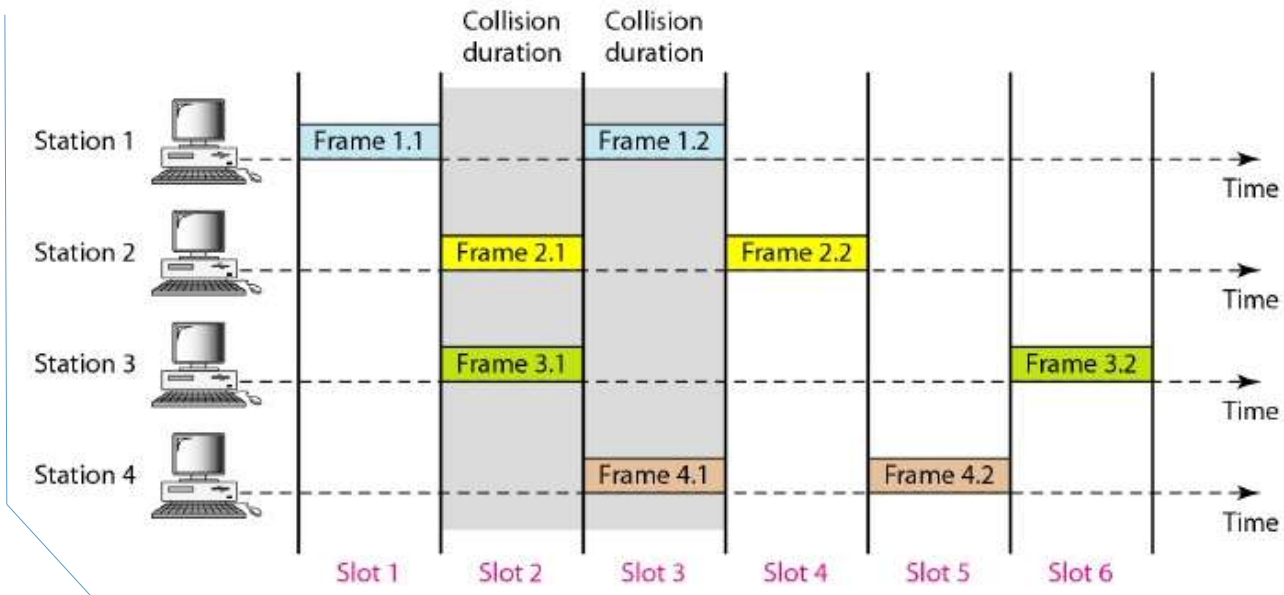


SLOTTED ALOHA

- Roberts proposed a mechanism for the betterment of ALOHA called as SLOTTED ALOHA
- In this mechanism, **frames are transmitted in fixed time slots**.
- **If any station is unable to send frame at the start of time slot, then it needs to wait until the start of next time slot**.
- Therefore, vulnerable period reduces from $2T_{fr}$ to T_{fr}
- **Different stations may send in the same time period (T_{fr}) thus overlapping occurs - not only small part of frame, but the entire frame**.

SLOTTED ALOHA





PERFORMANCE OF PURE & SLOTTED ALOHA

<u>Efficiency (Max. Throughput):</u>

- **Pure ALOHA (18.4%)**

$S = G * e^{-2G}$

G=0.5 → S=0.5/e = 0.184

Since, G=0.5=1/2

ie. half station sending data in T_{fr} time has max. throughput, which means for max. throughput, 1 station should get $2T_{fr}$ time

- **Slotted ALOHA (36.8%)**

$S = G * e^{-G}$

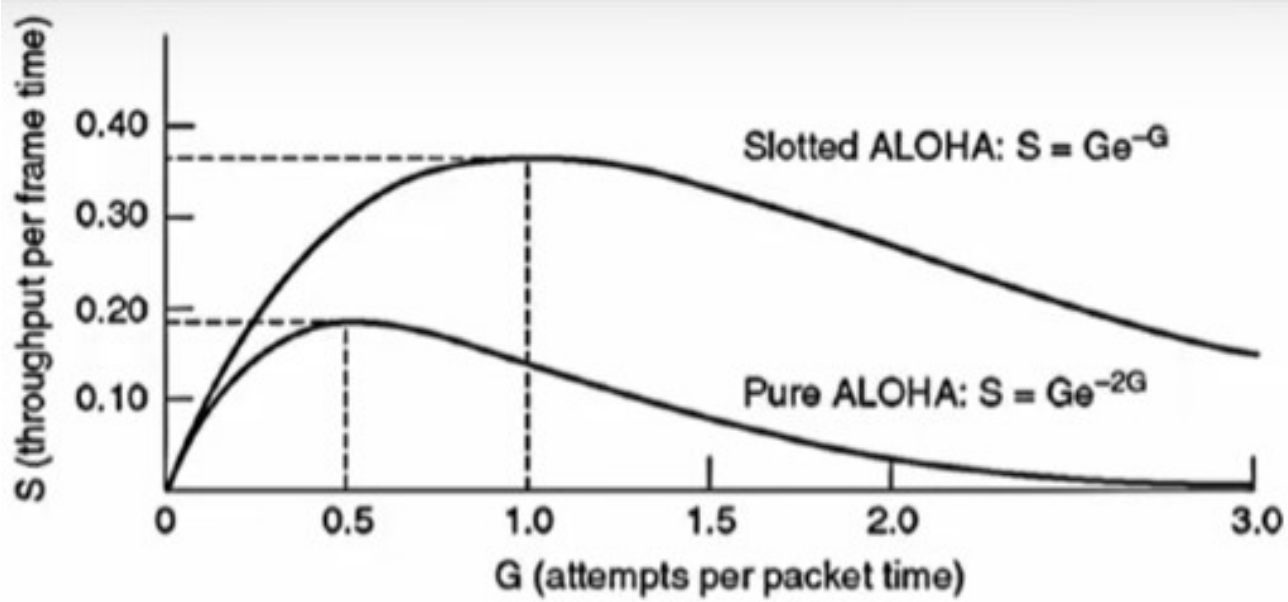
G=1→ S=1/e = 0.368

G → Offered Load

(No. of stations which want to transmit in T_t time)

S → Throughput

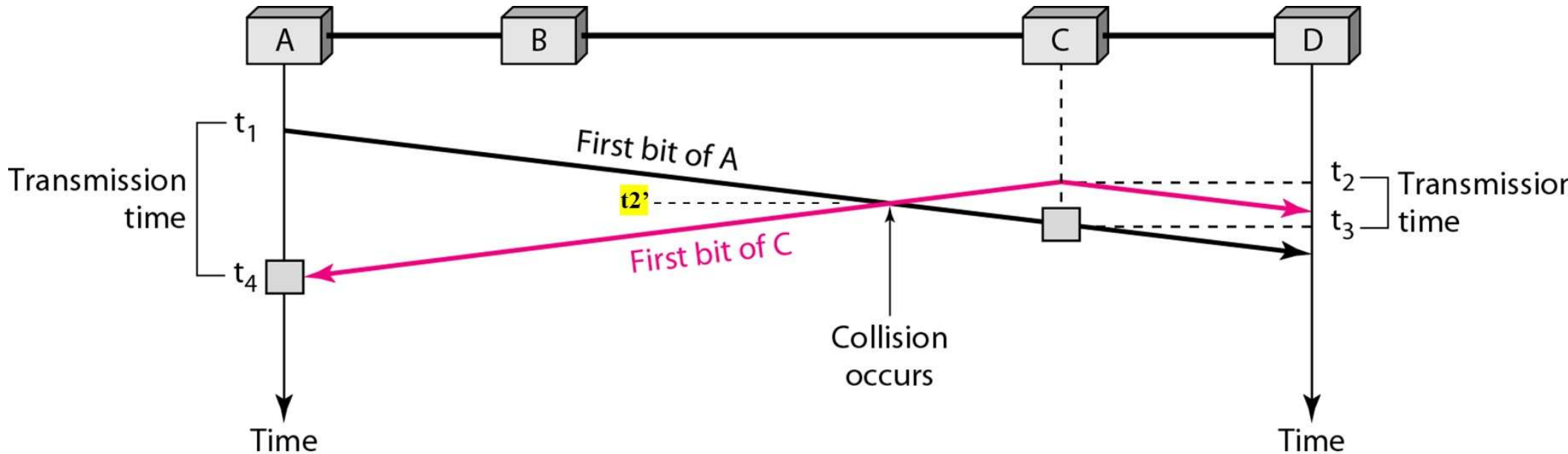
e → Euler's Number = 2.718 (Constant)



CSMA

- To minimize the chance of collision and, therefore, to increase the performance, the CSMA method was developed.
- The chance of collision can be reduced if a station **senses the medium before trying to use it**.
- Carrier sense multiple access (CSMA) requires that each station **first listen to the medium (or check the state of the medium) before sending**.
- CSMA is based on the principle - sense before transmit or "listen before talk" (LBT)
- **CSMA can reduce the possibility of collision, but it cannot eliminate it**.
- Stations are connected to a shared channel (usually a dedicated medium).
- The **possibility of collision still exists because of propagation delay**; when a station sends a frame, **it still takes time** (although very short) **for the first bit to reach every station and for every station to sense it**.
- In other words, **a station may sense the medium and find it idle,** only **because the first bit sent by another station has not yet been received**.

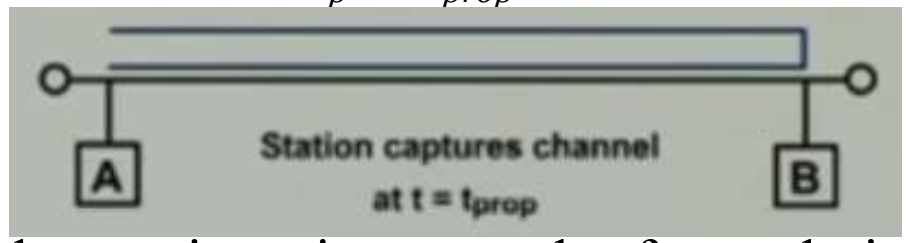
COLLISION IN CSMA – SPACE/TIME MODEL



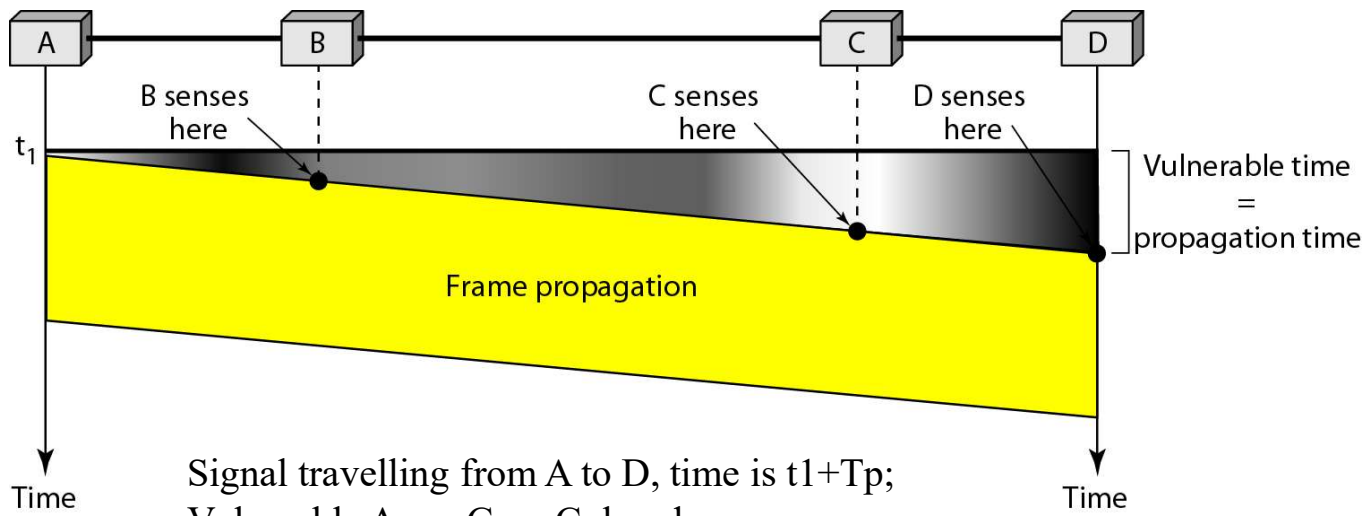
- **At time *t1* station A senses the medium and finds it idle**, so it sends a frame.
- **At time *t2 (t2> t1)* station C senses the medium and finds it idle** because, at this time, the first bits from station A have not reached station C.
- Station C also sends a frame.
- The two signals collide at time t2' and both frames get destroyed.

CSMA VULNERABLE PERIOD

- The vulnerable time for CSMA is the propagation time T_p *or* t_{prop}
- This is the time needed for a signal to reach (propagate) from one station (A) to the other station (B).



- When a station sends a frame, and any other station tries to send a frame during this time, a collision will result.
- But if the first bit of the frame reaches the end of the medium, every station will already have heard the bit and will refrain from sending.



Signal travelling from A to D, time is t1+Tp;
Vulnerable Area: Gray Colored

EFFICIENCY

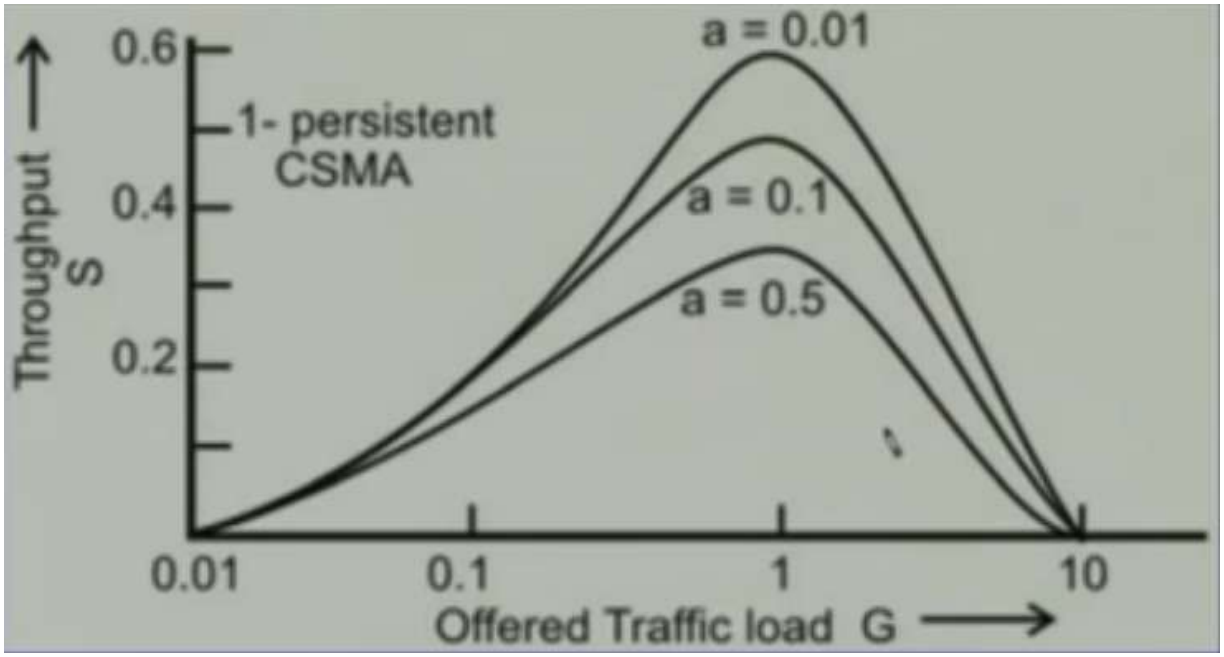
$$S = G / (1 + 2 * a)$$

S – Efficiency
G – Offered Load
a – Collission Probability

The Efficiency (*E*) is a measure of how effectively the network utilizes its available bandwidth, while the Throughput (*S*) is a measure of the actual data transfer rate.

Collission Probability (a):
a = propagation time/transmission time

Throughput becomes better if lesser is the propagation time and proportionately, more is the Transmission time.



PERSISTENCY

What should a station do if the channel is busy or idle?

1-persistent

- **If the medium is idle – transmit** (with probability of 1); else, ***continue to listen*** **until the channel is sensed idle**; then transmit immediately.
- Here is the highest chance of collision because two or more stations may find the line idle and send their frames immediately.

Non Persistent

- Sense the carrier, **if found busy –** ***wait for some random time*** and sense again, if found free – immediately send the data else wait and repeat above process.
- Here the chance of collision is reduced because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously.
- But, this method reduces the efficiency of the network because the medium remains idle when there may be stations with frames to send.

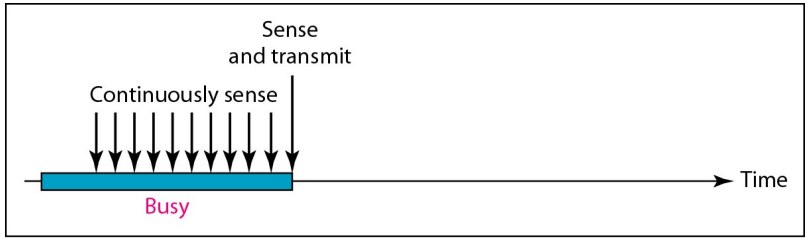
P-PERSISTENT

p-persistent

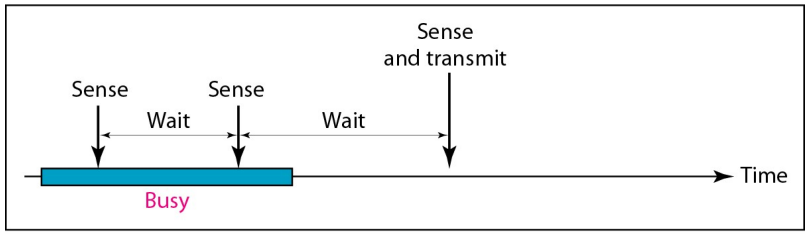
- It is used if the channel has time slots with a **slot duration equal to or greater than the maximum propagation time**.
- The p-persistent approach combines the advantages of the other two strategies.
- It reduces the chance of collision and improves efficiency.
- In this method, after the station finds the line idle it follows these steps:
  - ## With probability p, the station sends its frame.
  - With probability q = (1– p), **the station waits for the beginning of the next time slot and checks the line again**.
    - If the line is idle, it goes to step ##**.**
    - If the line is busy, it acts as though a collision has occurred and uses the back-off procedure.

p-send….q-wait….if busy…back-off…

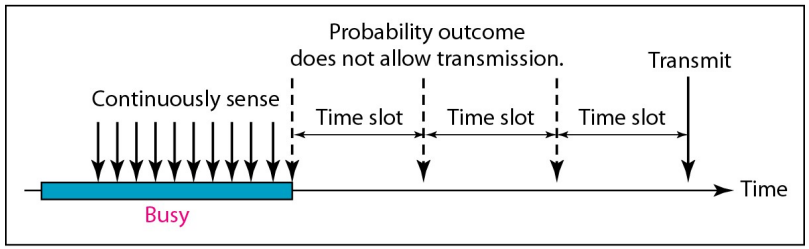
PERSISTENCY



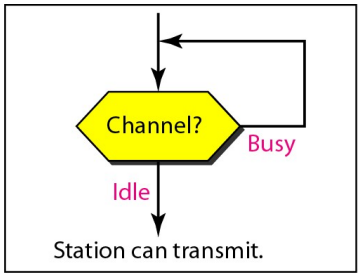
a. 1-persistent



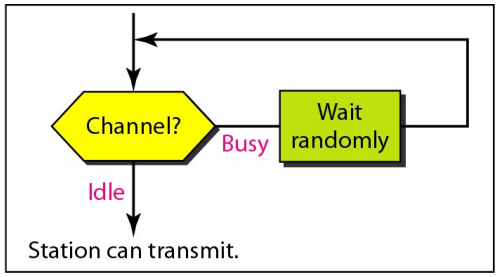
b. Nonpersistent



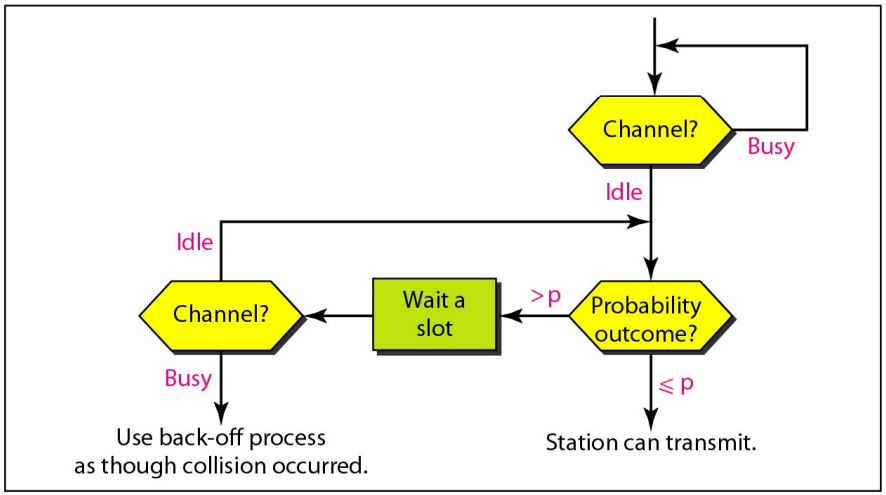
c. p-persistent



a. 1-persistent



b. Nonpersistent



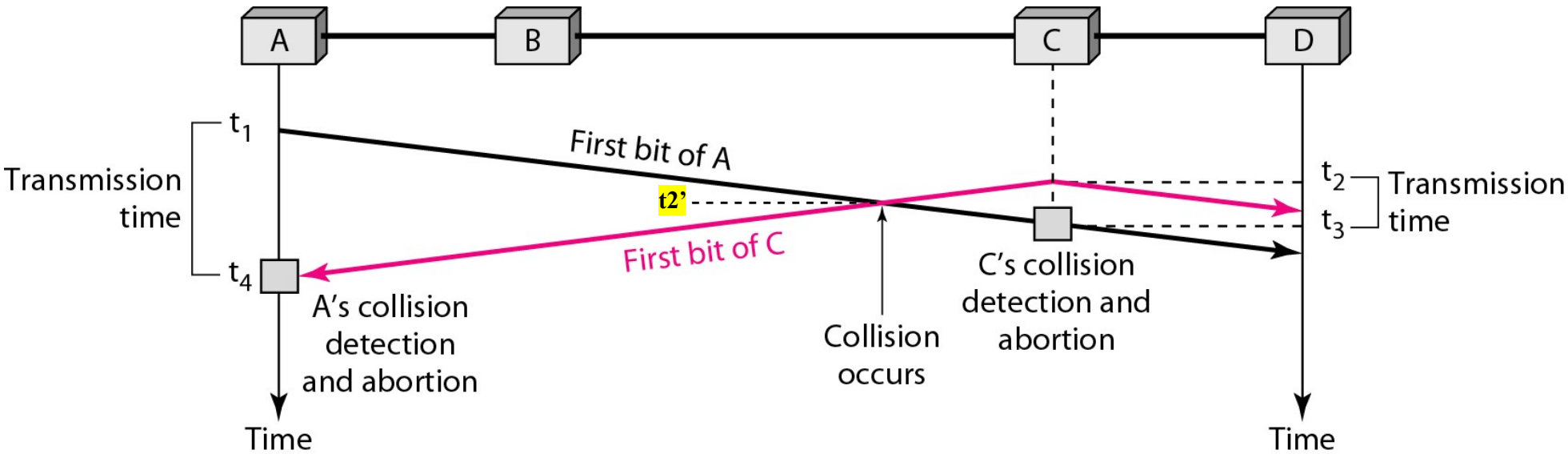
c. p-persistent

CSMA/CD

- The CSMA method does not specify the procedure following a collision. Carrier sense multiple access with collision detection (CSMA/CD) augments the algorithm to handle the collision.
- **Here, a station monitors the medium after it sends a frame to see if the transmission was successful. If a collision is found, the frame is sent again.**
- In CSMA/CD, a station **listens** to the medium **while transmitting** (LWT), if collision is found then retransmission is done.
- To better understand CSMA/CD, let us look at the first bits transmitted by the two stations involved in the collision. Although each station continues to send bits in the frame until it detects the collision, we show what happens as the first bits collide.

CSMA/CD – COLLISION OF FIRST BIT

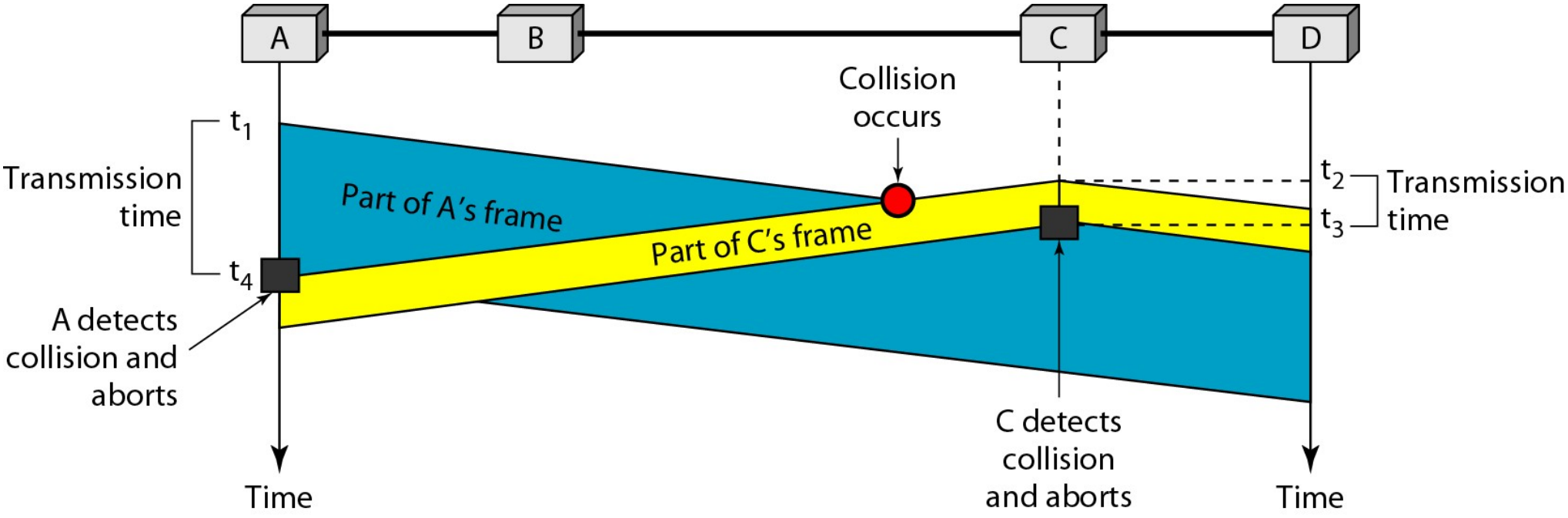
- At time *t*1, station A has executed its persistence procedure and starts sending the bits of its frame.
- At time *t2,* station C has not yet sensed the first bit sent by A.
- Station C executes its persistence procedure and starts sending the bits in its frame, which propagate both to the left and to the right.



CSMA/CD – COLLISION OF FIRST BIT

- The collision occurs sometime after time *t2, say t2'*
- Station C detects a collision at time *t3* when it receives the first bit of A's frame.
- Station C immediately (or after a short time, but we assume immediately) aborts transmission.
- Station A detects collision at time *t4* when it receives the first bit of C's frame; it also immediately aborts transmission.
- Looking at the figure, we see that A transmits for the duration *t4 – t1;*
- C transmits for the duration *t3* - *t2*.
- At time *t4,* the transmission of *A's* frame, though incomplete, is aborted;
- At time *t3,* the transmission of C's frame, though incomplete, is aborted.

CSMA/CD – COLLISON & ABORTION



A Complete Diagram

CSMA/CD - Minimum Frame Size

- For CSMA/CD to work, we need a restriction on the frame size.
- **Before sending the last bit of the frame, the sending station must check collision occurrence, if found, then abort the transmission**.
- This is so because – the station, once the entire frame is sent, does not keep a copy of the frame and does not monitor the line for collision detection.
- Therefore, the frame transmission time T_{fr} must be at least two times the maximum propagation time *Tp*.
- To understand the reason, let us think about the worst-case scenario:
  - If the two stations involved in a collision are the maximum distance apart, the signal from the first takes time *Tp* to reach the second, and the effect of the collision takes another time *Tp* to reach the first.
  - So the requirement is that the first station must still be transmitting after *2Tp* .

CSMA/CD - PROCEDURE

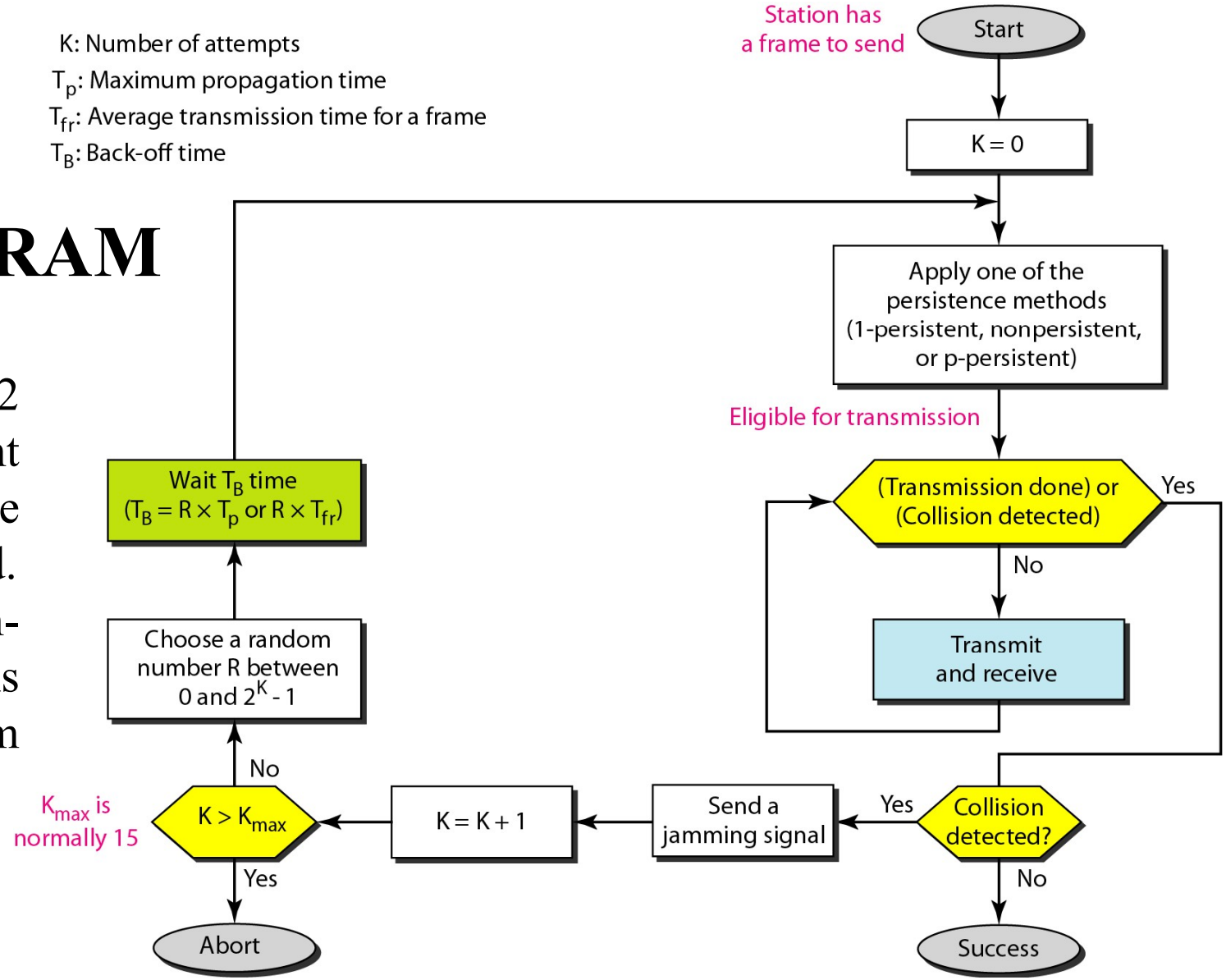
- Flow diagram for *CSMA/CD* is similar to the one for the ALOHA protocol, but there are some differences.
- The first difference is the addition of the persistence process.
- We need to sense the channel before we start sending the frame by using one of the persistence processes we discussed previously (nonpersistent, 1-persistent, or p-persistent).
- The second difference is the frame transmission.
- In ALOHA, we first transmit the entire frame and then wait for an acknowledgment.
- In *CSMA/CD,* transmission and collision detection is a continuous process.

CSMA/CD - PROCEDURE

- We do not send the entire frame and then look for a collision.
- The station transmits and receives continuously and simultaneously (using two different ports).
- We use a loop to show that transmission is a continuous process.
- We constantly monitor in order to detect one of two conditions: either transmission is finished, or a collision is detected.
- Either event stops transmission.
- When we come out of the loop, if a collision has not been detected, it means that transmission is complete; the entire frame is transmitted. Otherwise, a collision has occurred.
- The third difference is the sending of a short jamming signal that enforces the collision in case other stations have not yet sensed the collision.

CSMA/CD FLOW DIAGRAM

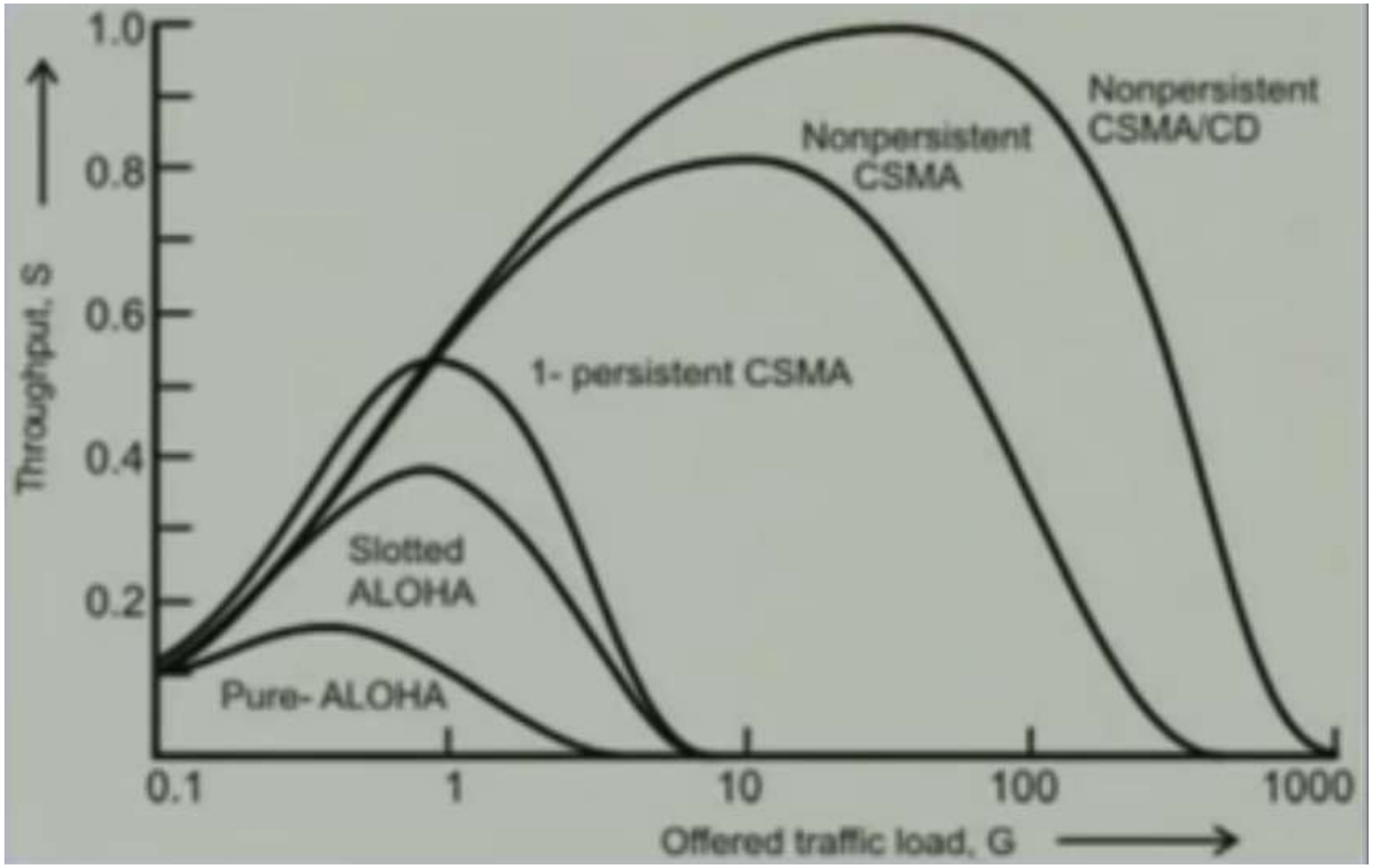
- Jamming signal (32 or 48) bits are sent whenever the collision is detected.
- It causes non-transmitting stations to wait until the jam signal ends



CSMA/CD – THROUGHPUT

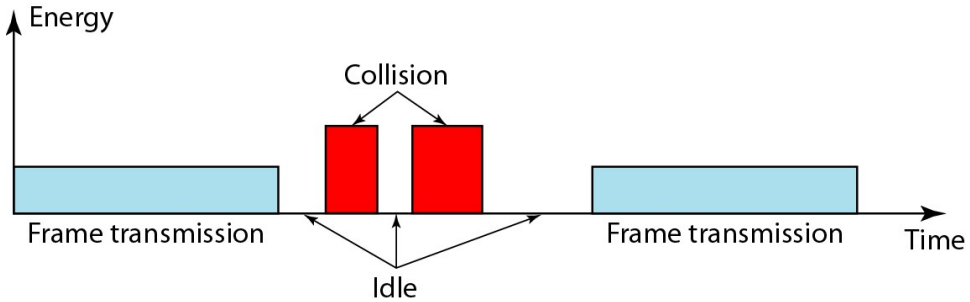
- The throughput of *CSMA/CD* is greater than that of pure or slotted ALOHA.
- The maximum throughput occurs at a different value of G and is based on the persistence method and the value of *p* in the p-persistent approach.
- For 1-persistent method the maximum throughput is around 50 percent when G =1.
- For non-persistent method, the maximum throughput can go up to 90 percent when G is between 3 and 8.

PERFORMANCE COMPARISON



CSMA/CD – ENERGY LEVEL

- Level of energy in a channel can have three values: **zero**, **normal**, and **abnormal**.
- At the **zero level**, the channel is idle.
- At the **normal level**, a station has successfully captured the channel and is sending its frame.
- At the **abnormal level**, there is a collision, and the level of the energy is twice the normal level.
- A station that has a frame to send or is sending a frame needs to monitor the energy level to determine if the channel is idle, busy, or in collision mode.
- Adjoining figure shows the situation.



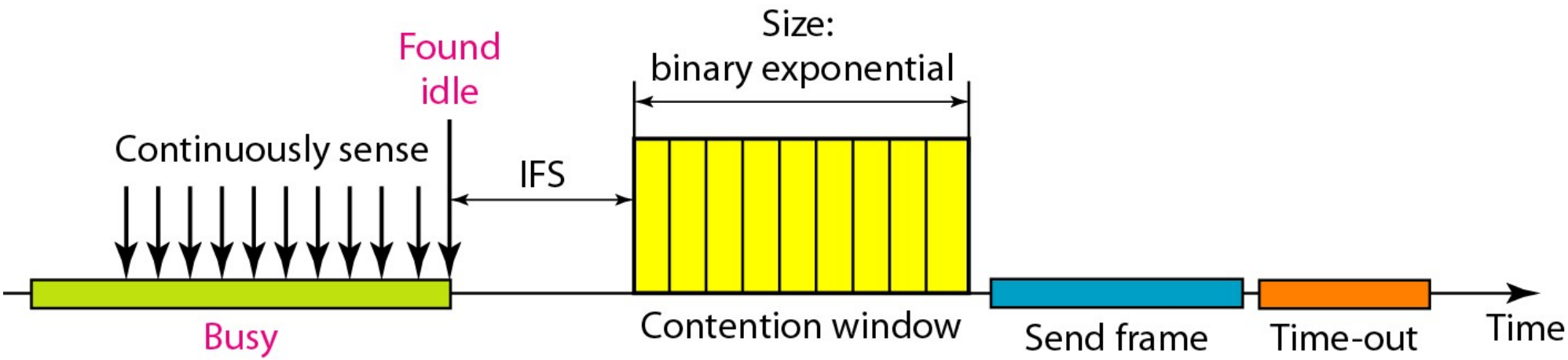
CSMA/CD ENERGY LEVEL

- The basic idea behind *CSMA/CD* is that a station needs to be able to receive while transmitting to detect a collision.
- When there is no collision, the station receives one signal: its own signal.
- When there is a collision, the station receives two signals: its own signal and the signal transmitted by a second station.
- To distinguish between these two cases, the received signals in these two cases must be significantly different. In other words, the signal from the second station needs to add a significant amount of energy to the one created by the first station.
- In a wired network, the received signal has almost the same energy as the sent signal because either the length of the cable is short or there are repeaters that amplify the energy between the sender and the receiver. This means that in a collision, the detected energy almost doubles.

CSMA/CA

- However, in a wireless network, much of the **sent energy is lost in transmission**.
- The received signal has very little energy. Therefore, a **collision may add only 5 to 10 percent additional energy**.
- This is not useful for effective collision detection.
- **We need to avoid collisions on wireless networks because they cannot be detected.**
- Carrier sense multiple access with collision avoidance *(CSMA/CA)* was invented for this network.
- Collisions are avoided through the use of CSMA/CA's three strategies: the **inter-frame space (IFS)**, the **contention window**, and **acknowledgments (ACK)**, as shown in next figure.

CSMA/CA – TIMINGS



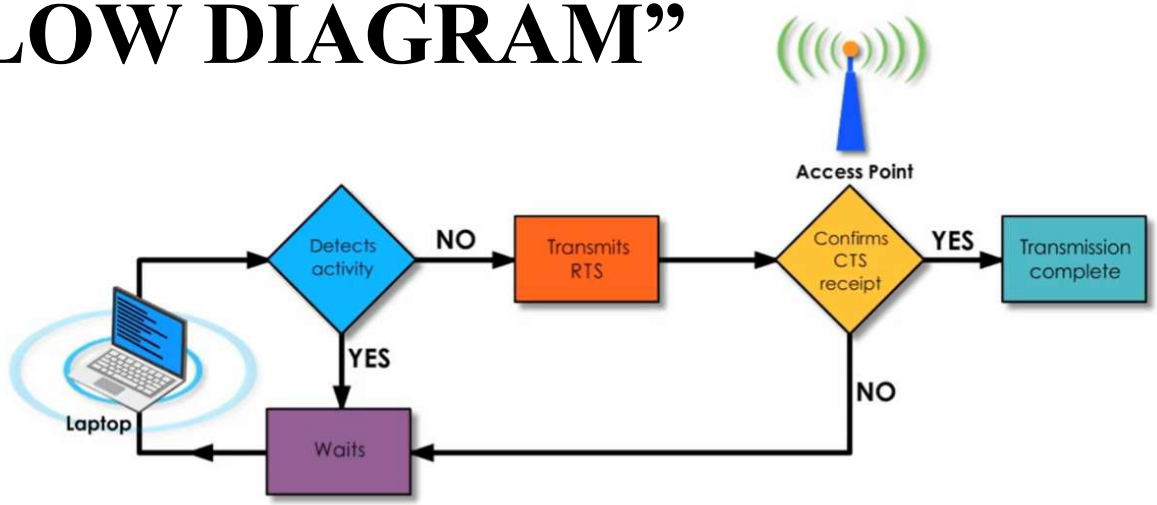
CSMA/CA – INTERFRAME SPACE (IFS)

- Firstly, collisions are avoided by deferring transmission even if the channel is found idle.
- When an idle channel is found, the station does not send immediately but waits for a period of time called the **inter-frame space or IFS**.
- Even though the channel may appear idle when it is sensed, a distant station may have already started transmitting.
- The distant station's signal has not yet reached this station.
- The IFS time allows the front of the transmitted signal by the distant station to reach this station.
- If after the IFS time the channel is still idle, the station can send, but it still needs to wait a time equal to the contention time (in previous slide).
- The IFS variable can also be used to prioritize stations or frame types.
  - eg. a station that is assigned a shorter IFS has a higher priority.

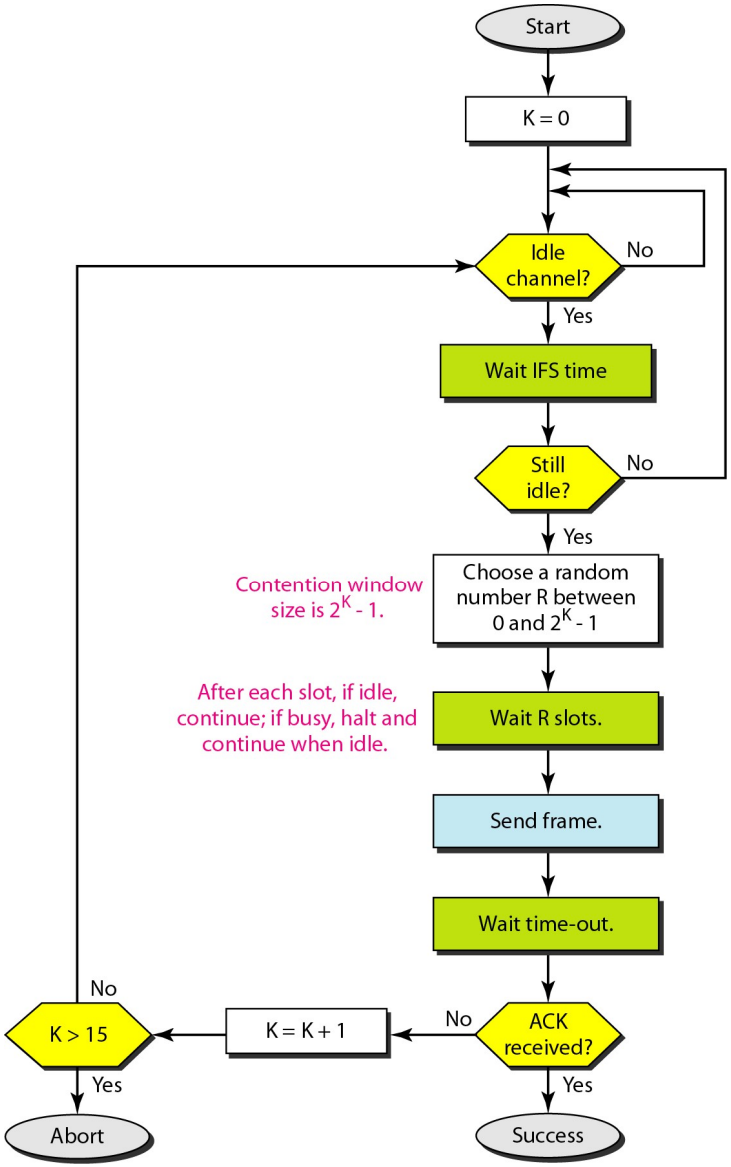
CSMA/CA – CONTENTION WINDOW

- The contention window is an amount of time divided into slots.
- A station that is ready to send chooses a random number of slots as its wait time.
- The number of slots in the window changes according to the binary exponential back-off strategy.
- This means that it is set to one slot - the first time and then doubles each time, the station cannot detect an idle channel after the IFS time.
- This is very similar to the p-persistent method except that a random outcome defines the number of slots taken by the waiting station.
- One interesting point about the contention window is that the station needs to sense the channel after each time slot.
- However, if the station finds the channel busy, it does not restart the process; it just stops the timer and restarts it when the channel is sensed as idle.
- This gives priority to the station with the longest waiting time.

CSMA/CA PROCEDURE "FLOW DIAGRAM"



- The channel needs to be sensed before and after the IFS.
- The channel also needs to be sensed during the contention time.
- For each time slot of the contention window, the channel is sensed.
  - If it is found idle, the timer continues;
  - If the channel is found busy, the timer is stopped and continues after the timer becomes idle again.



CSMA/CA – ACK & USAGE

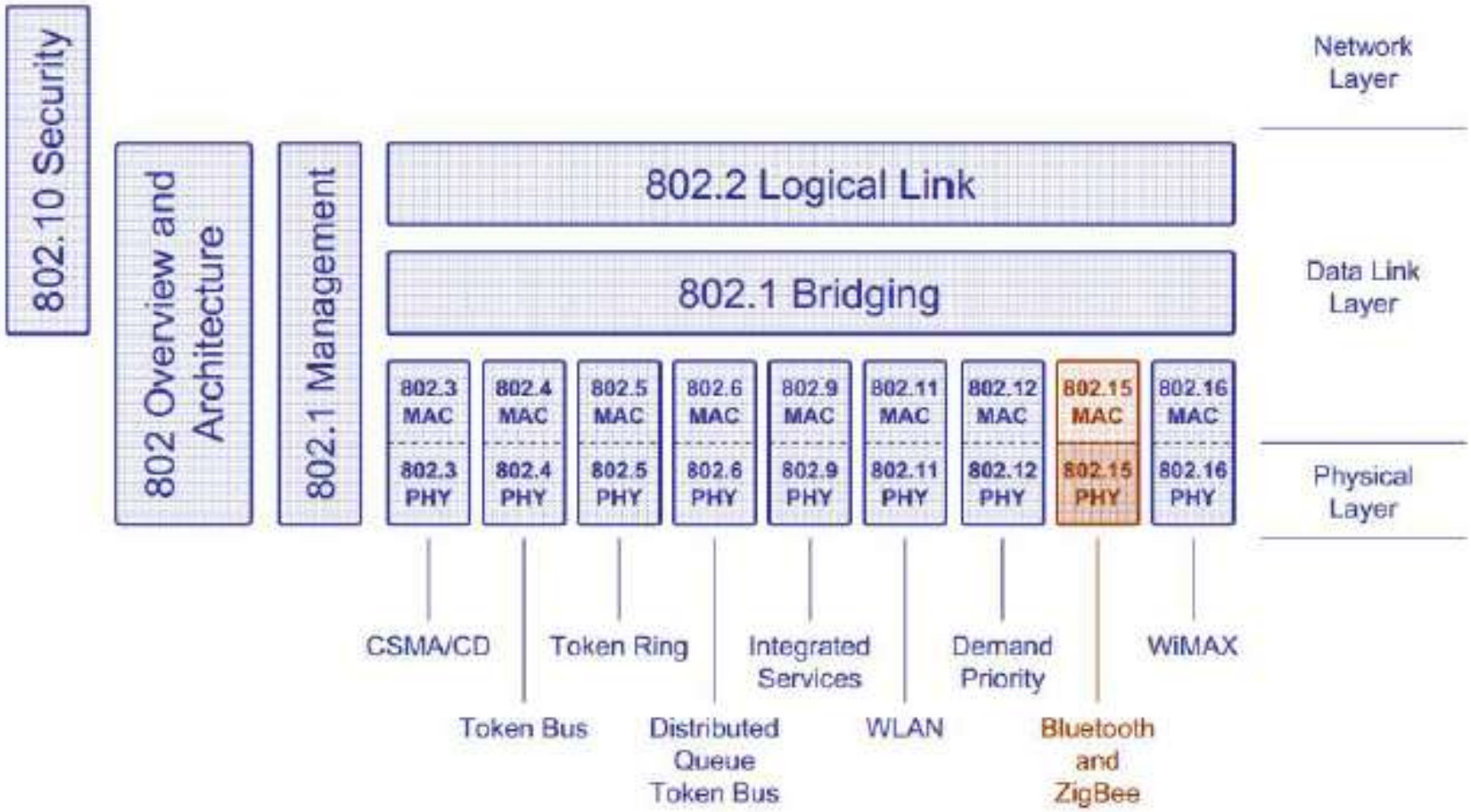
ACKNOWLEDGEMENT:

- With all these precautions, there still may be a collision resulting in destroyed data.
- In addition, the data may be corrupted during the transmission. The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.

FOR WIRELESS NETWORKS:

- CSMA/CA was mostly intended for use in wireless networks.
- The procedure described above, however, is not sophisticated enough to handle some particular issues related to wireless networks, such as hidden terminals or exposed terminals.
- These issues are solved by augmenting the above protocol with hand-shaking features.

IEEE 802

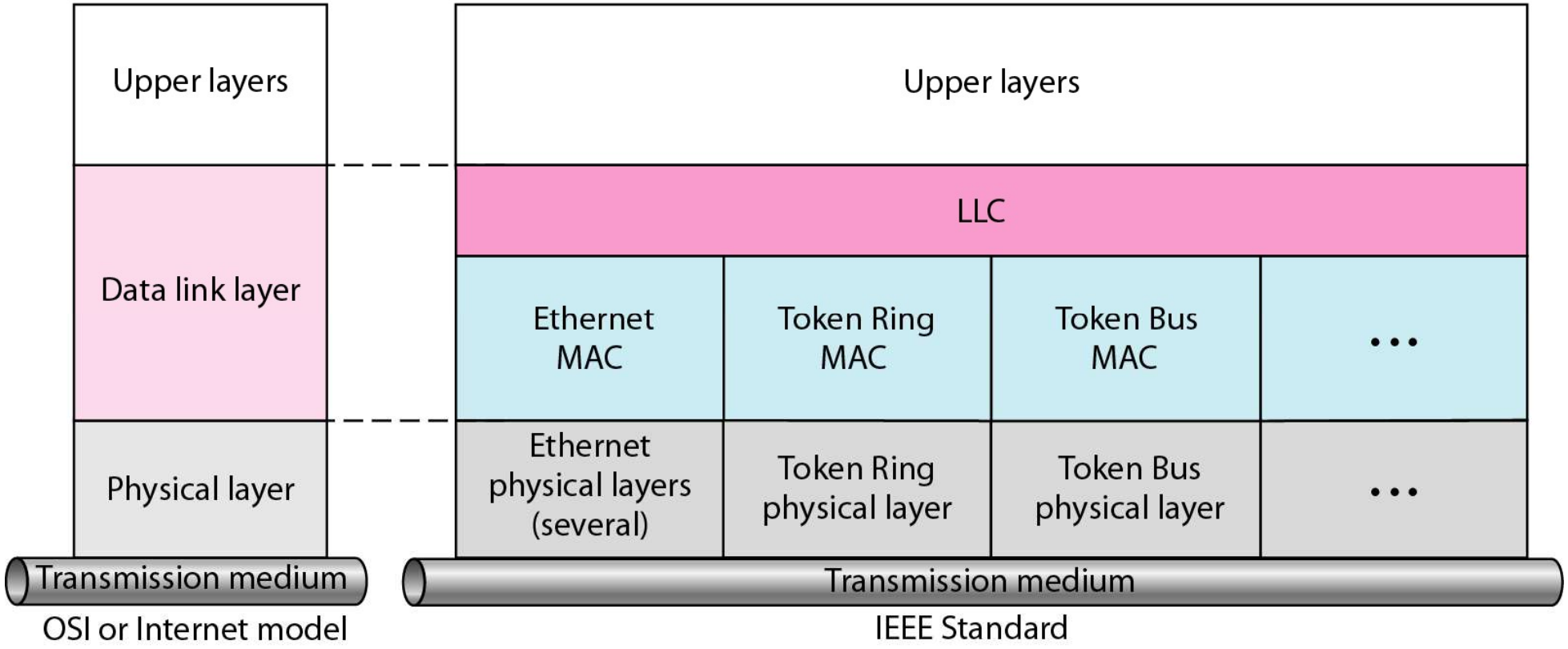

802.10 Security
802 Overview and Architecture
802.1 Management
802.2 Logical Link
802.1 Bridging
802.3 MAC
802.4 MAC
802.5 MAC
802.6 MAC
802.9 MAC
802.11 MAC
802.12 MAC
802.15 MAC
802.16 MAC
802.3 PHY
802.4 PHY
802.5 PHY
802.6 PHY
802.9 PHY
802.11 PHY
802.12 PHY
802.15 PHY
802.16 PHY
Network Layer
Data Link Layer
Physical Layer
CSMA/CD
Token Bus
Token Ring
Distributed Queue Token Bus
Integrated Services
WLAN
Demand Priority
Bluetooth and ZigBee
WiMAX

FUNCTIONS OF DATA LINK LAYER

- The data link layer transforms the physical layer, a raw transmission facility, to a link responsible for node-to-node (hop-to-hop) communication.
- It has two sublayers: LLC and MAC
- Specific responsibilities of the data link layer include:
  - Framing
  - Physical Addressing
  - Flow Control
  - Error Control
  - Media Access Control.

LLC & MAC PROTOCOL STACK

Single LLC protocol serving several MAC protocols



TRANSMISSION TIME

- It's the time taken by the node for transferring 1st bit of frame to last bit of same frame in the communication channel.
- Use Case:
  - A Node has a frame with 10,000 bits to transfer
  - Bandwidth of the medium is 10 Mbps
- Since,
  - Transmission Time = No. of bits to transfer / Bandwidth of the medium
    = 10,000 bits / 10*1,000*1000 bits per second
    = 1 milli second